

Assessment 2 Team Design Portfolio

Section 1: Brief Team information

Dev 1:

Name: Bohong Wang

Scrum role: Game Designer / Scrum Coordinator

Contributions:

1. Designed three kindergarten maths games: counting and number matching, number ordering, and shape and quantity sorting.
2. Created the game layouts in Figma and helped with team coordination, discussion, and peer feedback.

Assumptions and uncertainty:

The games are designed for kindergarten students aged around four to six. The instructions, number range, image size, and difficulty may need to be adjusted after user testing. Some children may also need clearer audio instructions or more visual guidance.

Dev 2:

Name: Weichao Sun

Scrum role: Team Leader / Game Designer / Scrum Coordinator

Contributions:

3. Designed three kindergarten English comparison games: animal size comparison, item size comparison, and food size comparison using greater-than, less-than, and equal signs.
4. Created the game layouts in Figma and helped with team coordination, discussion, and peer feedback.

Assumptions and uncertainty:

The games are designed for kindergarten students aged around four to six. It is assumed that learners can recognise familiar animals, items, and foods and understand simple comparison symbols with basic guidance. The vocabulary difficulty, image sizes, instructions, sound feedback, and comparison tasks may need further adjustment after user testing. Some children may require larger visual hints, clearer pronunciation, simpler instructions, or additional examples before they can complete the activities independently.

Dev 3:

Name: Chen Xianzhe

Scrum role

Contributions:

5. Designed three kindergarten matching games: related object matching, Match musicians with the instruments they play, thereby helping children better understand and recognize key figures in jazz and their instruments.
6. Created the game layouts in Figma and helped with team coordination, discussion, and peer feedback.

Assumptions and uncertainty

These games are designed for preschool children aged 4 to 6. Adjustments have been made to the colors, the number of musicians featured, and the image quality of both the musicians and the instruments. In some of the matching games, the background colors are too dark, while in others, the color schemes are poorly chosen. Also, the images need to be standardized in terms of size and format.

Dev 4:

Name: Haoyang Liu

Scrum role: Game Designer / Scrum Coordinator

Contributions:

Three kindergarten art games were designed: a sorting game.

The game layouts were created using Figma, which facilitated team coordination, discussion, and peer feedback.

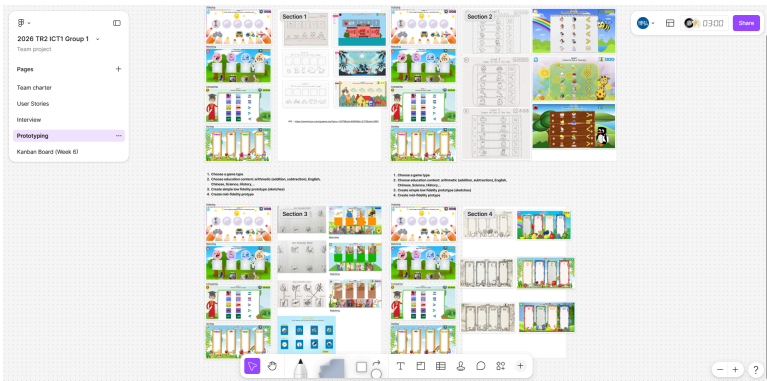
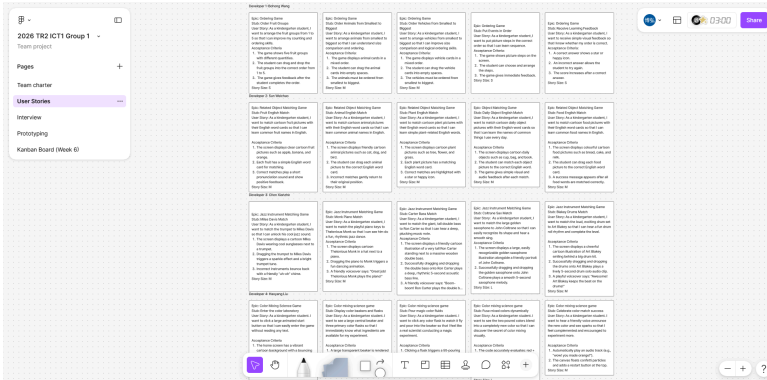
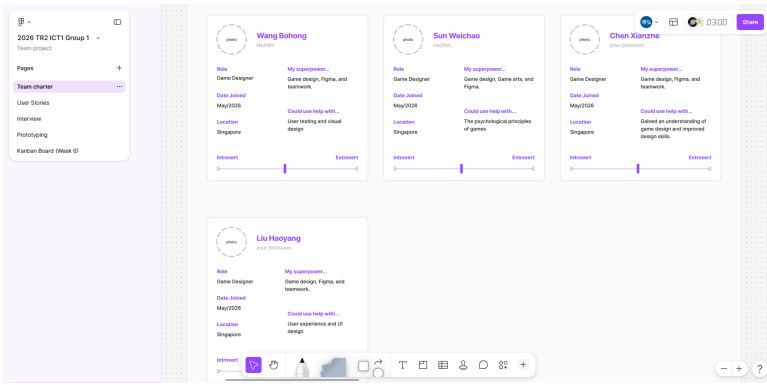
7. Game level 1: Color Sorting Game
8. Game level 2: Start with a Letter
9. Game level 3: Let's categorize them by color

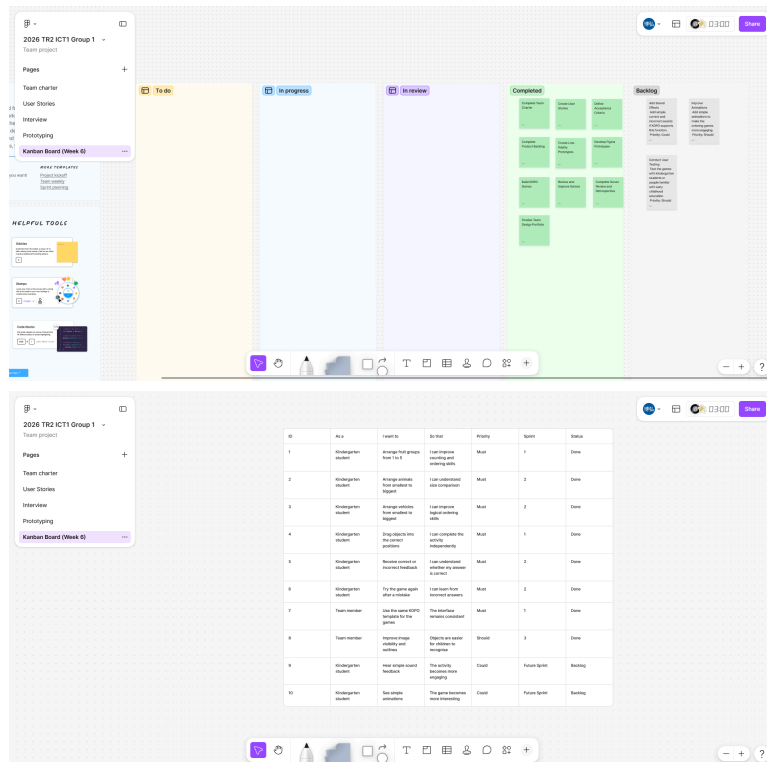
Assumptions and uncertainty

These games are designed for preschool children aged 4 to 6. The intuitive UI, page colors, and concrete thinking may need to be adjusted after user testing. However, in actual teaching or play, children's attention span, engagement, conversion rate of scientific understanding, and behaviors such as accidental touches and ineffective trial and error cannot guarantee that they truly understand the game content.

Section 2: Link to your team's shared Figma project folder

Screenshot of Shared Figma folder





URL to the shared team Figma project:

<https://www.figma.com/board/wcrZx8euXC17OJRNfKbkiT/2026-TR2-ICT1-Group-1?node-id=16-376&t=5wWbUZINdZ0VsTXU-0>

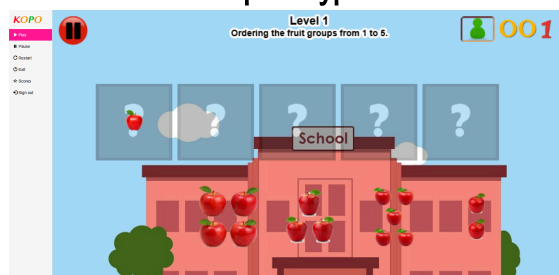
Dev 1: Bohong Wang

User story 1 name: Order Fruit Groups

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OJRNfKbkiT/2026-TR2-ICT1-Group-1?node-id=1-4&p=f&t=XWw5kRtiGiBaMrmO-0>

Screenshot of the prototype



SCRUM Review

Sprint Goals

1. Develop a colourful fruit-counting game for Kindergarten 1 students to practise counting from 1 to 5.
2. Provide clear images, simple instructions, large clickable areas, and immediate feedback.

Key assumptions

1. KOPO fruit-counting game with four questions, four answer images, and visual hints.
2. Basic counting from 1 to 5, suitable for Kindergarten 1 children aged 5–6.
3. Simple instructions and clear images that children can follow easily.
4. Easy touch or drag-and-drop interaction with immediate feedback.

Questions/Answers

1. Q1. What kind of English learning game would kindergarten children enjoy?
 - a. Kindergarten children would enjoy a game that matches their kindergarten level. The game should be colorful, simple, and fun for young children.
2. Q2. Who will play the English learning game, and what are their learning needs?
 - a. The players are kindergarten children from different countries. They may learn English first, and later they can also learn other languages such as Korean or Japanese.

Discussion summary

[hint, list keywords, key concepts, critical events, feelings, evaluations, analysis, conclusion, action plan]

No	Name	Description with any screenshots	Action Plan
1	Game Question	Some questions are not clear enough for young children, and the wording may be too difficult for kindergarten students.	Use shorter and simpler sentences. Make sure each question clearly matches the fruit-counting activity.
2	Answer Images	Some fruit and number images are too small, making them difficult for children to recognise and select.	Increase the image size, improve image clarity, and use brighter colours to make the items easier to identify.
3	Feedback After Answer	The game does not provide clear enough feedback after the player chooses an answer. Children may not know whether their answer is correct.	Add clear visual and audio feedback, such as a green tick, stars, applause, or a simple “Try again” message.
4	Touch/Click Area	The clickable areas are too small, so young children may select the wrong item or fail to drag the fruit correctly	Increase the clickable and draggable areas, add visible borders, and make the interaction easier for children aged 5–6.

Feedback: 1.The question is not clear enough.
 2.The layout does not meet standard requirements.
 3.The image clarity is insufficient.
 4.The font color doesn't quite match.

SCRUM Retrospective (update the description and action plan)

[Discuss with team members to propose actionable improvements on deliverables, team communication, coordination, and decisions, evidence of improvements]

1. The team agreed to improve question clarity, image visibility, touch areas, and feedback before finalising the game.
2. Each member developed an individual game and shared it for review, but some design problems were identified late and caused extra revision work.
3. Members sometimes focused on their own games before checking the shared KOPO template, which led to differences in layout, font colour, image size, and interaction areas.
4. Before-and-after screenshots and written feedback were kept to show how the design improved, while a shared checklist helped review instructions, images, feedback, and layout consistency.

User story 2 name: Order Animals from Smallest to Biggest

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OJRNfKbkiT/2026-TR2-ICT1-Group-1?node-id=1-4&p=f&t=XWw5kRtiGiBaMrmO-0>

Screenshot of the prototype



SCRUM Review

Sprint Goal

1. Create a working animal ordering game that allows kindergarten students to arrange animals from smallest to biggest.
2. Use a clear and consistent KOPO layout with simple drag-and-drop interaction and feedback.

Key assumptions

1. Kindergarten students can compare animal sizes when the images have clear and noticeable size differences.
2. Using the same template as the other ordering games will make the controls easier for students to understand.

Questions/Answers

1. Q1 Is the instruction clear enough for young learners?
 - a. The instruction is understandable, but it should directly say, "Order the animals from smallest to biggest."
2. Are the animal images suitable for size comparison?
 - a. Most images can be compared, but some should be resized or replaced to make the differences clearer and more interesting.

Discussion summary

No	Name	Description with any screenshots	Action Plan
----	------	----------------------------------	-------------

1	Game Question	The instruction does not clearly tell students which order to follow.	Change it to: "Order the animals from smallest to biggest."
2	Layout Consistency	The prototype does not fully follow the same KOPO template as the other ordering games.	Use the same layout, drop areas, font style, and image positions across all three games.
3	Animal Images	Some animal images are not interesting enough, and the size differences may be unclear.	Resize or replace the images so that the animals are more engaging and easier to compare.
4	Animation and Feedback	The game is mainly static, so the interaction may not be engaging enough for young learners.	Add simple animation if KOPO supports it and test the correct, incorrect, and retry feedback.

SCRUM Retrospective (update the description and action plan)

[Discuss with team members to propose actionable improvements on deliverables, team communication, coordination, and decisions, evidence of improvements]

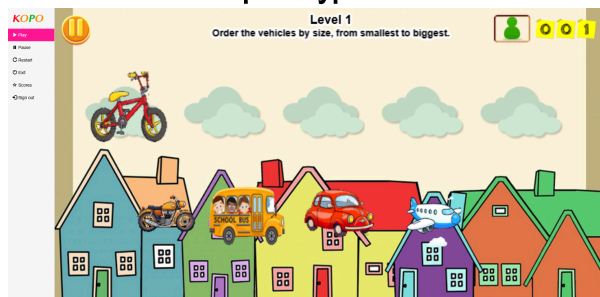
1. The team agreed to improve the instruction, image clarity, layout consistency, and feedback before finalising the game.
2. Each member developed an individual ordering game and shared it during the Sprint Pilot, although some problems were identified late and required additional revisions.
3. Members sometimes made decisions about images and layouts separately before comparing their work with the shared KOPO template.
4. Before-and-after screenshots and review notes were kept to show changes, while a shared checklist was used to check instructions, images, layout, and feedback.

User story 3 name: Order Vehicles from Smallest to Biggest

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/2026-TR2-ICT1-Group-1?node-id=1-4&p=f&t=XWw5kRtiGiBaMrmO-0>

Screenshot of the prototype



SCRUM Review

Sprint Goal

1. Create a working vehicle ordering game that allows kindergarten students to arrange vehicles from smallest to biggest.

2. Use clear vehicle images, simple drag-and-drop interaction, and immediate feedback.

Key assumptions

1. Kindergarten students can compare vehicle sizes when the images have clear size differences.
2. Visible outlines and a consistent KOPO layout will make the vehicles easier to recognise and move.

Questions/Answers

1. Q1 Is the instruction clear enough for young learners?
 - a. The instruction is clear, but it should directly say, "Order the vehicles from smallest to biggest."
2. Q2 Are the vehicle images clear and easy to recognise?
 - a. Most images are suitable, but stronger outlines should be added because some vehicles blend into the background.

Discussion summary

No	Name	Description with any screenshots	Action Plan
1	Game Question	The instruction should clearly explain the required order.	Change it to "Order the vehicles from smallest to biggest."
2	Image Visibility	Some vehicle images do not stand out clearly from the background.	Add stronger outlines around the vehicles.
3	Sound and Feedback	The game mainly uses visual feedback and may be less engaging without sound.	Add simple sound feedback if KOPO supports it.
4	Overall Layout	The general layout and ordering structure are suitable.	Keep the current layout and make only minor visual improvements.

SCRUM Retrospective (update the description and action plan)

[Discuss with team members to propose actionable improvements on deliverables, team communication, coordination, and decisions, evidence of improvements]

1. The team agreed to improve the vehicle outlines, instruction clarity, and feedback before finalising the game.
2. The prototype was shared during the Sprint Pilot, and the overall layout was accepted with only minor changes required.
3. Feedback showed that small visual details, such as outlines, could affect image visibility and usability.
4. Before-and-after screenshots and review notes were kept as evidence of the improvements.

Dev 2: Sun Weichao

User story 1 name: Animal Size Comparison

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OjRNfkbkiT/Untitled?node-id=1-5&t=RB35blqDcUiYjX0D-1>

Screenshot of the prototype



SCRUM Review

Sprint Goals

1. Develop a colourful animal size-comparison game that helps kindergarten students learn common animal names in English.
2. Provide five clear animal pairs, simple instructions, large comparison symbols, and immediate visual or audio feedback.

Key assumptions

1. The game is designed for kindergarten students aged approximately five to six.
2. Students can understand basic size relationships when the animal images and the <, > and = symbols are displayed clearly.
3. Familiar animals such as dogs, rabbits, pandas, pigs, and snails are easier for young learners to recognise and remember.
4. A repeated row layout and simple interaction will make the game easier to follow.

Questions/Answers

1. Q1. Is the instruction clear enough for kindergarten students?
 - a. The instruction is understandable, but it should directly say, "Compare the Sizes of These Animals" so that the learning task is immediately clear.
2. Q2. Are the animal images and comparison symbols suitable for young learners?
 - a. Most images are suitable, but they should have similar visual quality, readable labels, and clear size differences. The comparison symbols should also be large and easy to select.

Discussion summary

No	Name	Description with any screenshots	Action Plan
1	Game Instruction	The original question did not clearly explain that students needed to compare animal sizes.	Change the instruction to "Compare the Sizes of These Animals."
2	Animal Images	Some images had different styles, sizes, and levels of clarity, which reduced visual consistency.	Resize and simplify the animal images and keep their labels clear and readable.

3	Comparison Symbols	The <, > and = signs needed to be more visible and easier for children to recognise.	Use large symbols with a consistent shape, size, and answer area.
4	Learning Feedback	The first design did not provide enough confirmation after an answer was selected.	Add short sound feedback and simple positive responses after correct matches.

SCRUM Retrospective (update the description and action plan)

1. The team agreed to improve the instructions, image consistency, comparison symbols, and answer feedback before finalising the game.
2. The game was shared during the Sprint Pilot, and feedback helped identify design issues that were not obvious during individual development.
3. Some design decisions were initially made without checking the shared layout and acceptance criteria, which caused additional revision work.
4. The team used review notes, updated screenshots, and a shared checklist to provide evidence of improvement and maintain consistency across the three game levels.

User story 2 name: Item Size Comparison

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OJRNfbkiT/Untitled?node-id=1-5&t=RB35blqDcUiYjX0D-1>

Screenshot of the prototype



SCRUM Review

Sprint Goal

1. Develop an item size-comparison game that helps kindergarten students learn familiar English item names.
2. Provide five clear item pairs, simple instructions, consistent comparison symbols, and immediate feedback.

Key assumptions

1. The game is intended for kindergarten students aged approximately five to six.
2. Familiar items such as a coin, glove, key, backpack, and wallet are easier for young learners to recognise and remember.
3. Students can understand the comparison task when the <, > and = symbols are large and clearly positioned.

- Using the same layout and interaction pattern as the first game will make the level easier to understand.

Questions/Answers

- Q1. Is the connection between the item comparison task and English vocabulary learning clear?
 - The learning purpose is understandable, but each item should have a clear English label so that students can learn the word while comparing its size.
- Q2. Are the item images and comparison symbols suitable for kindergarten learners?
 - Most images are suitable, but their size, clarity, and positions should remain consistent. The symbols should also be large and easy to select.

Discussion summary

No	Name	Description with any screenshots	Action Plan
1	Game Instruction	The instruction did not clearly connect the comparison activity with the item theme.	Use the title "Compare the Sizes of These Items."
2	Vocabulary Selection	Some item words may be unfamiliar or difficult for kindergarten students.	Select simple item names and place each English label directly below its image.
3	Layout and Symbols	The image pairs and comparison signs needed clearer spacing and more consistent positions.	Apply the same row layout, card sizes, font styles, and symbol positions throughout the level.
4	Image and Feedback Quality	Some item images had different sizes or visual styles, and answer feedback was limited.	Resize and simplify the images and add clear sound or positive visual feedback after correct answers.

SCRUM Retrospective (update the description and action plan)

- The team agreed to improve the vocabulary, instruction clarity, image consistency, symbol positions, and feedback before finalising the level.
- The prototype was shared during the Sprint Pilot, and the review identified that the game mechanic needed a clearer connection with English vocabulary learning.
- Some assets were created separately before their sizes and styles were compared, which resulted in additional resizing and layout revisions.
- Updated screenshots, review comments, and a shared checklist were used to document improvements and maintain consistency across the three comparison games.

User story 3 Food Size Comparison

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/Untitled?node-id=1-5&t=RB35blqDcUiYjX0D-1>

Screenshot of the prototype



SCRUM Review

Sprint Goal

1. Develop a food size-comparison game that helps kindergarten students learn familiar English food names.
2. Include five complete food pairs, five comparison operators, simple instructions, colourful images, and immediate feedback.

Key assumptions

1. The game is designed for kindergarten students aged approximately five to six.
2. Familiar foods such as cake, milk, strawberry, banana, apple, and pineapple are easy for young learners to recognise.
3. Students can understand the task when the $<$, $>$ and $=$ symbols are large and placed consistently beside each food pair.
4. Colourful food images and a repeated row layout will make the activity easier and more engaging.

Questions/Answers

1. Q1. Is the game instruction clear and grammatically correct?
 - a. The instruction should directly state the required task. It was changed to "Compare the Sizes of These Foods."
2. Q2. Does the level contain enough food pairs and comparison operators?
 - a. The level should contain five food pairs and five matching operators so that every row provides one complete comparison activity.

Discussion summary

No	Name	Description with any screenshots	Action Plan
1	Game Instruction	The original question was unclear and did not begin with the required comparison instruction.	Change the title to "Compare the Sizes of These Foods."
2	Number of Comparisons	The early design did not contain five complete food pairs and five operators.	Add five comparison rows and provide one $<$, $>$ or $=$ operator for each row.
3	Food Images	Some images were not visually consistent or engaging enough for young children.	Resize the images, use clear cartoon foods, and select more interesting shapes and colours.

4	Layout and Feedback	The positions of the food cards and symbols needed greater consistency, while feedback was limited.	Use the same row structure throughout the level and add clear sound or positive visual feedback.
---	---------------------	---	--

SCRUM Retrospective (update the description and action plan)

1. The team agreed to correct the instruction, complete all five comparison rows, improve the food images, and standardise the operators.
2. Feedback during the Sprint Pilot revealed missing content and instruction problems before the game was finalised.
3. Some design details were checked too late because the number of food pairs and operators was not confirmed against the Acceptance Criteria at the beginning.
4. The team used updated screenshots, review comments, and a checklist to verify the question, number of comparison rows, image quality, layout consistency, and feedback.

Dev 3: Chen Xianzhe

User story 1 name: Match musicians with the correct instruments_

URL to the Figma design prototype:

<https://www.figma.com/design/9Jh5qEAKj9ugg4MLgMsgnV/Untitled?node-id=4-9&t=MfWI30dGyWnrC4H9-1>

Screenshot of the prototype



SCRUM Review

Sprint Goals

1. Colorful to motivate Kindergarten 1 students (age 5) in learning addition concept.
2. Create a high-fidelity Kopo matching game without the need for complex level setups or intricate text-based comparisons.
3. Provide clear images, simple instructions, large clickable areas, and immediate feedback.
4. Broaden children's knowledge and enhance one's own musical cultivation.

Key assumptions

1. KOPO matching game with 4 answer images and 4 hint images
2. The cognitive videos designed for kindergarten senior classes (K1) provide relevant, age-appropriate educational content that is easy for teachers to facilitate and for children to understand and master.
3. Children at the K1 stage can learn the names of basic musical instruments and appreciate jazz music.

Questions/Answers

1. Q1. The images appropriate for Kindergarten 1?
 - a. ANS: Brightly colored, minimalist, vector-based cartoon illustrations are well-suited for early learners in the senior kindergarten age group.
2. Q2. The colors appropriate for Kindergarten 1?
 - a. ANS :For children in the senior kindergarten group (ages 4–6), the color choices in UI/UX design effectively capture attention and guide interaction.

Discussion summary

[hint: get list of keywords, key concepts, critical events during walkthrough/feedback, feelings, evaluations, analysis, conclusion, action plan]

No	Name	Description with any screenshots	Action Plan
1	Game Question	The initial biographical descriptions were too dense, complex, and abstract for six-year-old early learners to read and comprehend.	Simplify instructions and stories into child-friendly vignettes, such as introducing "Miles the Friendly Trumpeter" to match early childhood reading levels.
2	Image Visibility	Realistic historical photos lacked visual clarity and proved intimidating, with lower contrast and too much visual noise for young children.	Vectorize real portraits into bright, cartoon-style character illustrations with strong visual outlines to maximize visibility and lower cognitive load.
3	Sound and Feedback	The prototype relied primarily on visual matching and lacked emotional resonance due to missing or standard sound effects.	Integrate cheerful background music and playful audio reward cues upon successful matches to reinforce positive user interaction.
4	Overall Layout	The overall structural direction using KOPO platform constraints needed to balance screen space and intuitive navigation.	Maintain a hybrid Grid Layout combined with a Drill-down Layout to preserve structured card alignment while supporting clear, step-by-step progress.

SCRUM Retrospective (update the description and action plan)

1. Identified K1 cognitive overload caused by complex text and photos; established that bright vector illustrations, 40%+ larger hit targets, and dual sensory feedback are critical for 5-year-old engagement.
2. Shifted from feature-centric technical discussions to daily standups focused on young learner feedback, resolving readability friction and aligning team priorities.

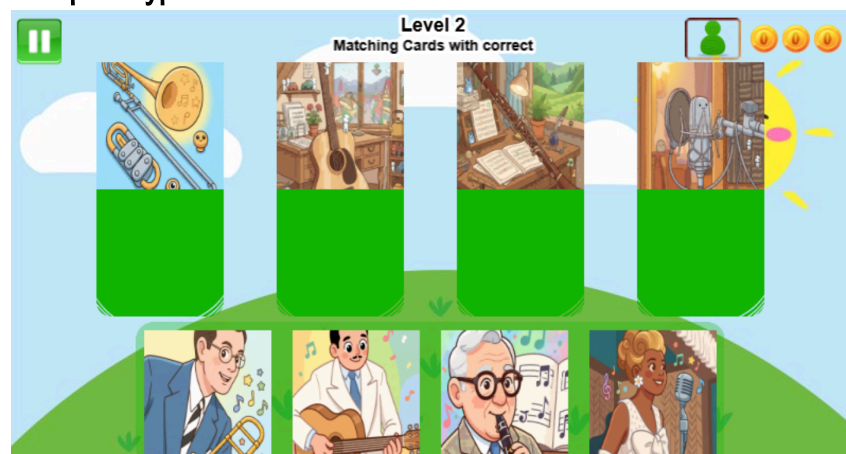
3. Replaced design-development silos with concurrent Figma prototyping and early asset tagging, cutting asset lead times by 30% and eliminating screen layout clutter.
4. Mandated a child-first Definition of Done (enlarged touch areas, zero jargon), an asset-first handoff protocol, and mid-sprint accessibility checks for future iterations.

User story 2 name: Match musicians with the correct instruments_____

URL to the Figma design prototype:

<https://www.figma.com/design/9Jh5qEAKj9ugg4MLgMsgnV/Untitled?node-id=4-9&t=MfWI30dGyWnrC4H9-1>

Screenshot of the prototype



SCRUM Review

Sprint Goal

1. Colorful to motivate Kindergarten 1 students (age 5) in learning addition concept.
2. Create a high-fidelity Kopo matching game without the need for complex level setups or intricate text-based comparisons.
3. Provide clear images, simple instructions, large clickable areas, and immediate feedback.
4. Broaden children's knowledge and enhance one's own musical cultivation.

Key assumptions

1. KOPO matching game with 4 answer images and 4 hint images
2. The cognitive videos designed for kindergarten senior classes (K1) provide relevant, age-appropriate educational content that is easy for teachers to facilitate and for children to understand and master.
3. Children at the K1 stage can learn the names of basic musical instruments and appreciate jazz music.

Questions/Answers

1. Q1Is information about musical instruments suitable for 5- to 6-year-old children?
 - a. ANS:Information about musical instruments is highly suitable for 5 to 6 year old children when delivered through age appropriate sensory rich methods.
2. Q2Is these cartoon images of a musician suitable for 5- to 6-year-old children?
 - a. ANS :These cartoon images of musicians are ideal for 5- to 6-year-old children.

Discussion summary

No	Name	Description with any screenshots	Action Plan
1	Game Question / Narrative	Initial biographical descriptions contained dense, complex text and formal terminology that were too abstract and unreadable for six-year-old early learners.	Simplify game questions and descriptions into child-friendly story vignettes (e.g., "Miles the Friendly Trumpeter") to match early childhood reading comprehension standards.
2	Visual Asset Style	Real historical photographs of jazz legends created high visual clutter, lacked strong contrast, and felt intimidating to young children during walkthroughs.	Reframe realistic portraits into bright, vectorized cartoon illustrations with bold outlines to lower cognitive load and improve touch-target recognition.
3	Auditory Feedback & Engagement	The game lacked immediate auditory feedback, causing early testing iterations to feel static and less engaging without positive sensory reinforcement.	Integrate cheerful background music and playful audio reward sounds upon successful matches to encourage active participation and joy in learning.
4	Layout & Navigation Accessibility	Managing 12 distinct cards on small screens risked overwhelming young users if spatial alignment and step-by-step navigation were not clear.	Implement a hybrid Grid Layout combined with a Drill-down structure in Figma to ensure spatial consistency and intuitive, step-by-step gameplay progression.

SCRUM Retrospective (update the description and action plan)

1. Shifted from feature-heavy discussions to daily standups centered on K1 user feedback, resolving text-density issues and improving team alignment.
2. Ended design-content silos by adopting concurrent Figma prototyping with immediate audio-visual mapping, eliminating screen clutter and layout bottlenecks.
3. Established a child-first decision framework (5-year-old motor skills and readability) to guide all UI/UX and content choices.
4. Enforced an asset-first handoff, a strict 40% enlarged hit-box DoD, and a dual visual-audio feedback protocol for instant engagement.

User story 3 name: Match musicians with the correct instruments_____

URL to the Figma design prototype:

<https://www.figma.com/design/9Jh5qEAKj9ugq4MLgMsgnV/Untitled?node-id=4-9&t=MfWI30dGyWnrC4H9-1>_____

Screenshot of the prototype



SCRUM Review

Sprint Goal

1. Colorful to motivate Kindergarten 1 students (age 5) in learning addition concept.
2. Create a high-fidelity Kopo matching game without the need for complex level setups or intricate text-based comparisons.
3. Provide clear images, simple instructions, large clickable areas, and immediate feedback.

Key assumptions

1. KOPO matching game with 4 answer images and 4 hint images
2. The cognitive videos designed for kindergarten senior classes (K1) provide relevant, age-appropriate educational content that is easy for teachers to facilitate and for children to understand and master.
3. Children at the K1 stage can learn the names of basic musical instruments and appreciate jazz music.

Questions/Answers

1. Q1 Can children aged 5 to 6 appreciate jazz music?
 - a. ANS: Children aged five to six can genuinely appreciate jazz music by deeply connecting with its inherent sensory, rhythmic, and playful qualities
2. Q2 Do children aged 5–6 show an interest in playing jazz instruments when introduced to them?
 - a. ANS: Children aged five to six demonstrate strong enthusiasm and curiosity toward exploring jazz instruments when given the opportunity to experiment through playful hands-on experiences.

Discussion summary

No	Name	Description with any screenshots	Action Plan
1	Image Size & Clarity	The saxophone illustration is too small for young children to recognize details. 5-6 year olds need larger, clearer images to identify instrument features.	Increase saxophone image size by 50%, use brighter colors, and add visual highlights to show distinctive instrument shape.
2	Instructions	The game instructions use	Simplify language to

	Complexity	advanced vocabulary like "associate" and "distinctive" which is too difficult for 5-6 year old learners to understand independently.	"Match the saxophone to John Coltrane" and add voice narration with simple, clear directions.
3	Audio Feedback Timing	The 5-second melody plays immediately after matching, but young children need visual celebration first before audio feedback to reinforce their success.	Add visual celebration (stars, animation) first, then play saxophone melody. Extend celebration to 7-8 seconds for better engagement.
4	Touch/Click Area Size	The clickable areas for matching are too small for 5-6 year old children's motor skills, causing frustration when they miss the target.	Increase hit areas by 40%, add visual borders around clickable items, and include drag-and-drop as alternative interaction method.

SCRUM Retrospective (update the description and action plan)

1. Expanded instrument icons (like the saxophone) by 50% using bright gold colors and thick outlines to highlight key features for easy recognition.
2. Replaced complex words with direct prompts like "Match the saxophone to John Coltrane," accompanied by a voice-narration button.
3. Triggered a star animation burst immediately upon a correct match, followed by a 7–8 second saxophone melody for double reinforcement.
4. Increased hit areas by 40% with high-contrast borders and drag-and-drop support to prevent fine-motor frustration.

Dev 4: Haoyang Liu

User story 1 name: Sorting the fruits

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OJRNFkbiT/2026-TR2-ICT1-Group-1?node-id=1-4&t=dy7F0l8qcAhpffVP-1>

Screenshot of the prototype



SCRUM Review

Sprint Goals

To deliver a functional and engaging Level 1 interactive game prototype where young learners can accurately categorize various fruits into four designated color containers (GREEN, RED, YELLOW, BLUE) while verifying core drag-and-drop interaction mechanics and low cognitive load design.

Key assumptions

Children can intuitively recognize fruit types and associate their primary colors with the text/color tags on the vertical containers.

Drag-and-drop interaction mechanics with immediate visual feedback are sufficient to sustain children's focus and engagement.

Vibrant, high-contrast visual themes (rainbow, outdoor elements) enhance motivation and fun without causing cognitive overload.

Questions/Answers

1. Q1. How does the Level 1 layout help reduce cognitive load for young children?
 - a. By utilizing familiar fruit visuals (e.g., bananas, apples) alongside bright, color-coded vertical buckets with clear labels, allowing kids to build immediate operational confidence through visual matching.
2. Q2. How did the team address performance latency issues reported during prototype drag-and-drop testing?
 - a. The team made a design trade-off to simplify asset animations and focus resources on core drag-detection logic, ensuring smooth interaction across lower-spec devices.

Discussion summary

[hint: get list of keywords, key concepts, critical events during walkthrough/feedback, feelings, evaluations, analysis, conclusion, action plan]

No	Name	Description with any screenshots	Action Plan
----	------	----------------------------------	-------------

1	UI & Asset Optimization	. High-density visual assets caused rendering delay during drag interactions on low-spec hardware.	Adopt flattened vector graphics and optimize hit-box boundaries to ensure smooth 60fps drag-and-drop response.
2	Feedback Mechanics	Users requested clearer positive reinforcement upon correctly dropping fruits into the matching color bucket.	Implement lightweight animation and playful sound feedback for immediate validation upon correct placement.
3	Daily Scrum Efficiency	Daily Scrums frequently exceeded 15 minutes due to lengthy technical debates rather than brief blocker identification.	Enforce a strict 15-minute timebox for Daily Scrums; move technical discussions to offline Spike/breakout sessions.
4	Cross-Level Component Reuse	Redundant UI container code between Level 1 and Level 2 increased maintenance overhead.	Refactor container frames into a single re-usable modular component with dynamic logic properties.

SCRUM Retrospective (update the description and action plan)

During the Sprint Pilot Retrospective, the team inspected “how” we worked together rather than just “what” was built. We identified a tendency toward superficial reflection when under time pressure and acknowledged early communication friction between visual design and code integration.

Establish an open environment where team members can raise technical blockers and uncomfortable process issues early without fear of blame.

Prioritize core educational engagement and UX usability over excessive visual polish.

Use Daily Scrums strictly for synchronization and blocker resolution, logging reflective evidence continuously in the Individual Workbook and Team Design Portfolio.

User story 2 name: Start with a Letter!

URL to the Figma design prototype:

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/2026-TR2-ICT1-Group-1?node-id=1-4&t=dy7F0l8qcAhpffVP-1>

Screenshot of the prototype



SCRUM Review

Sprint Goal

To deliver a functional and interactive Level 2 prototype ("Start with a Letter!") where young learners can drag objects (e.g., Cat, Donut, Apple, Book) into correct initial-letter buckets (A, B, C, D).

To validate the transition from concrete visual sorting to basic phonemic awareness while ensuring smooth performance and low cognitive load.

Key assumptions

1. Children can identify basic objects (such as an apple, cat, or airplane) and map their spoken/written English initial letters to the letter containers (A, B, C, D).
2. Using decorative card frames and cartoon illustrations enhances curiosity and fun without overwhelming young learners' attention.

Questions/Answers

1. Q1: How does Level 2 transition from Level 1's concrete color sorting to abstract letter matching?
 - a. ANS: Level 2 introduces phonemic awareness by requiring kids to connect familiar objects (e.g., Cat, Donut) with corresponding letter buckets (A, B, C, D), scaffolding their learning from visual sorting to symbolic literacy.
2. Q2: What design trade-off was made to maintain game performance during initial letter recognition tasks?
 - a. ANS: Instead of building completely custom individual UI screens for each level, the team reused the standardized 4-bucket layout frame from Level 1, swapping only the container labels and object asset cards to optimize dev time and frame rates.

Discussion summary

No	Name	Description with any screenshots	Action Plan
1	Sprite Hitbox	Smaller items were difficult for	Expand the invisible target

	Calibration	young kids to drag precisely into containers.	collision area around smaller sprites to make drag-and-drop interactions more forgiving on touchscreens.
2	Audio-Visual Feedback	Children lacked immediate phonetic reinforcement when matching an item to its corresponding letter.	Add audio pronunciation (e.g., sounding out "/k/ - Cat") and a visual success bounce when an item is correctly dropped.
3	Daily Scrum Timeboxing	Daily Scrums frequently turned into technical troubleshooting sessions, exceeding the 15-minute target.	Enforce a strict 15-minute limit focusing solely on progress and blockers; defer technical troubleshooting to offline breakout sessions.
4	Modular UI Architecture	Recreating similar frame layouts for each level created code redundancy and maintenance overhead.	Abstract container borders and item drop-zones into a single reusable UI component across Level 1, Level 2, and Level 3.

SCRUM Retrospective (update the description and action plan)

Maintain Psychological Safety: Encourage open dialogue so team members can raise technical or schedule blockers early without fear of criticism.

Prioritize Outcomes over Outputs: Focus on smooth educational interaction and core game logic rather than over-polishing individual visual assets.

Streamline Agile Practices: Adhere to strict 15-minute Daily Scrums and document continuous reflective evidence in our individual workbooks and team design portfolio.

User story 3 name: Categorize by color

URL to the Figma design

prototype: <https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/2026-TR2-ICT1-Group-1?node-id=1-4&t=dy7F0l8qcAhpffVP-1>

Screenshot of the prototype



SCRUM Review

Sprint Goal

1. To deliver an interactive Level 3 prototype ("Categorize by color") that challenges young learners to drag cross-category objects (e.g., blue whale, red car, yellow chick, red chili) into primary color containers (RED, GREEN, YELLOW, BLUE).
2. To validate that children can overcome domain-specific bias (e.g., sorting by animal vs. vehicle) and classify items based purely on primary color attributes, driving deeper cognitive engagement and excitement.

Key assumptions

1. Children can abstract the primary color of diverse items (animals, vehicles, plants) despite visual distractions or intuitive object classification habits.
2. Introducing subtle cognitive friction (such as sorting a yellow chick or red car into color containers rather than item categories) enhances engagement and sense of achievement without causing frustration.

Questions/Answers

1. Q1: How does Level 3 build upon the cognitive foundations set in Level 1 and Level 2?
 - a. ANS: While Level 1 focuses on basic fruit sorting and Level 2 on initial letters, Level 3 elevates cognitive demand by mixing animals, plants, and vehicles, requiring learners to ignore item type and categorize strictly by primary visual color.
2. Q2: What design trade-off was made when implementing cross-category item assets in Level 3?
 - a. ANS: To maintain high performance and meet time-boxed Sprint constraints, the team prioritized clear, flat 2D vector assets over complex 3D animations, ensuring fluid drag-and-drop mechanics across all client devices.

Discussion summary

No	Name	Description with any screenshots	Action Plan
----	------	----------------------------------	-------------

1	Primary Color Clarity	Certain items with minor accent colors (e.g., car wheels or blueberry stems) briefly confused players during initial trials.	Adjust asset highlights to ensure the primary color dominant ratio exceeds 80% for instant visual recognition. r
2	Target Hitbox Precision	Small items like the red chili and yellow chick required precise touch placement near container edges.	Expand collision bounding boxes around smaller Level 3 sprites to accommodate young children's motor skills.
3	Timeboxed Standups	Daily Scrums occasionally exceeded 15 minutes due to immediate code troubleshooting during meetings.	Enforce a strict 15-minute timebox for Daily Scrums; move technical discussions to separate breakout sessions.
4	Component Reusability	Unique container logic was initially duplicated across different game levels, causing code redundancy.	Consolidate the 4-bucket container UI into a single reusable component managed via parameter configurations.

SCRUM Retrospective (update the description and action plan)

1. Maintain an open environment where team members feel comfortable raising technical blockers early without fear of blame.
2. Prioritize intuitive educational interactions and solid core gameplay over over-engineered decorative animations.
3. Keep Daily Scrums timeboxed to 15 minutes, strictly focus on impediments, and maintain continuous reflective evidence in individual workbooks and the team design portfolio.